Information security – Tutorial 4

- 1 Enhancing Physical Security Controls

The table below is based on my Tutorial 1 table. I have added more Physical Security controls to improve the table.

Physical security	Technical security	Operational security
Locked doors	Passwrods	Rules and policies
ID cards	Multi-facto authentication	Staff training
Cameras	Firewalls	Backup plans
Security guards	Antivirus software	Incident response plans
Alarm systems	Encryption	Risk management
Lockedserver rooms	VPN	Change management
Fences	Access control system	Acceptable use policies

- 2 Identifying rabilities in Physical Security Controls

Locked doors :	Doors can be left open or not locked properly.
ID cards :	ID cards can be lost, stolen, or shared with others.
Cameras :	Cameras may not work, may not record, or may not be monitored.
Security guards :	Guards can make mistakes or miss suspicious activity.
Alarm systems :	Alarms may not be tested or may be ignored.
Locked server rooms :	Doors can be propped open or accessed by people who do not need access.
Fences :	Fences can be climbed, cut, or damaged.
Laptop locks :	Laptop locks can be broken or not used by employees.
Biometric access :	Biometric systems can fail or be bypassed if not managed correctly.

Security lighting :	Lights can stop working, leaving dark areas around the building.

These vulnerabilities show that physical controls are useful, but they can still fail if they are not managed, checked, and maintained.

- 3 Strengthening Weak Physical Security Controls

Locked doors :	Check doors regularly and make sure they close and lock properly.
ID cards :	Disable lost or stolen cards immediately and review access regularly.
Cameras :	Test cameras often and make sure recordings are stored safely.
Security guards :	Give guards proper training and clear procedures.
Alarm systems :	Test alarms regularly and make sure someone responds to them.
Locked server rooms :	Use access logs and only allow authorized staff inside.
Fences :	Inspect fences regularly and repair damage quickly.
Laptop locks :	Train staff to use laptop locks and store laptops safely.
Biometric access :	Combine biometric access with ID cards or PIN codes.
Security lighting :	Check lights regularly and replace broken lights quickly.

These improvements make the physical security controls stronger and reduce the chance of unauthorized access, theft, or damage.

- 4 Understanding Assets and Their Protection

An asset is anything that has value to an organization. This can include people, computers, IT systems, networks, software, data, buildings, and physical equipment. NIST defines an asset as anything that has value to an organization, including people, computing devices, IT systems, software, networks, and related hardware.

Locked doors :	Buildings, offices, equipment, and people
ID cards :	Restricted areas, employees, and company property
Cameras :	People, buildings, equipment, and evidence after incidents
Security guards :	Employees, visitors, buildings, and equipment
Alarm systems :	Buildings, offices, and equipment
Locked server rooms :	Servers, data, network equipment, and IT systems
Fences :	Building perimeter, parking areas, and outdoor equipment
Laptop locks :	Laptops, files, and company data
Biometric access :	High-security rooms, server rooms, and sensitive systems
Security lighting :	Outdoor areas, entrances, employees, and visitors

Physical security controls protect both physical and digital assets. For example, a locked server room protects physical servers, but it also protects the data stored on those servers.

- 6 Real-World Example

One real-world example is a university data center security issue. A recent audit of Eastern Connecticut State University found that sensitive IT areas were not properly secured. The report mentioned physical security problems such as doors being propped open and people being able to access areas without a clear need. The university responded by improving electronic door access, access reviews, training, and documentation.

Incident :	Sensitive IT areas were not properly secured.
Physical asset:	Data center, network closets, servers, and IT equipment

Vulnerability:	Doors were propped open and access controls were bypassed.
Failed control:	Physical locks, card readers, and access control procedures
Consequence:	Increased risk to business operations and sensitive IT systems
Improvement:	Better electronic access control, access reviews, staff training, and documentation

This example shows that physical security is important for information security. If someone can physically access a server room or IT area, they may be able to damage systems, steal equipment, or access sensitive data.

Physical security is an important part of information security. Controls such as locked doors, ID cards, cameras, guards, and locked server rooms help protect valuable assets. However, these controls can have weaknesses. To make them stronger, organizations should test them, maintain them, train staff, and review access regularly. Physical security protects both physical assets and information assets.